

An obstruction theoretic description of integrated information

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Abstract

We define the integrated cohomology groups of physical systems, and show that integrated information can be detected by the non-vanishing of H^1 obstruction classes that measures the failure of the cause-effect properties of the system to decompose into smaller parts. This allows us to further define higher order integration in a very natural way, and gives a completely functorial notion of integrated information, as suggested by Tull–Kleiner.

1 Introduction

Integrated Information Theory (IIT) is a mathematical description of the existing consciousness of a system, developed mainly by Giulio Tononi and his collaborators, [CITE](#). The theory is centered around the computation of Φ , which is a measure of the *integrated information* in the system. Informally, a system S in a state s , has integrated information, captured by $\Phi(S, s) > 0$, if the information generated by the entire system’s cause-effect structure in state s is greater than the information generated by the combined cause-effect structure of partitions of the system. The formal mathematical framework for computing Φ is rigorous and technical, and uses detailed features of probability theory and measure theory, see [CITE](#).

In the present paper we wish to study integrated information through the lens of obstruction theory, which is a mathematical framework studying certain obstruction classes that measures whether a property can be extended through defined levels of sub-structure. They live in mathematical objects called *cohomology groups*, which are abstract formulations of the amount of holes a system has in different dimensions. Obstruction theory was originally developed by Stiefel and Whitney to prove the non-existence of certain vector-fields on manifolds [CITE](#), but has later been applied in a plethora of mathematical contexts [CITE](#).

The conceptual similarity between these two ideas is also extendable to a formal description, which is the central aim of this short paper. Building on work by Kleiner and Tull on a categorical description of integrated information theory, see [CITE](#), we assign a simplicial complex $N^\bullet$ to an IIT system S , and use this to build cohomology groups valued in a sheaf $\mathcal{F}$ that incorporates the cause-effect dynamics of S . Intuitively, we wish to build a mathematical structure whose n -dimensional holes represent n -th order integration, or causal connectedness. We then define a natural obstruction class Δ^ψ for any attempted partition ψ of S into subsystems, which we show implies properties for IIT. In particular, if there is a non-trivial obstruction class, then the system has integrated information. These classes will detect whether the global cause-effect dynamics of S can be “glued together” from the local

dynamics of its decompositions. This gluing is described by the sheaf, and the failure to glue is measured by the cohomology.

Theorem 1.1. *Let s be a state of a system S . If the obstruction classes $\Delta^\Psi \in H^0(N^\bullet; \mathcal{F})$ vanish, then $\Phi(S, s) > 0$.*

The upshot of this work is that one can study integrated information from a new lens, and apply fundamental ideas from algebraic topology and homological algebra to study phenomena in consciousness. In particular, our theory gives a much more general approach to integrated information, that side-steps a lot of technicalities. We expect that our methods can be combined with topological data analysis, see [CITE](#) for an introduction, to locate minimum partitions (MIP) and connect integrated information to new developments in the relationship between topology and information theory, see for example [CITE](#) and [CITE](#). Note that we do not consider the size of Φ in this paper, delegating such an understanding in this framework to future work.

We make no claim surrounding the validity of IIT to explain consciousness and the properties of experience, and make no contributions to the ongoing debates [CITE](#). We only wish to connect some of IIT's mathematical properties to other mathematical concepts known to measure similar features. We view this work not as a new standard for IIT, but as a starting point to explore some of the rich mathematical structures related to it.

2 IIT systems

In [CITE](#), Tull–Kleiner develop a category theoretic foundation for integrated information theory, based on symmetric monoidal categories with some extra structure that allows for the modeling of cause-effect structures. We will use this as a backdrop for our cohomology theory, hence we quickly review their setup.

2.1 Process theories and decompositions

A *process theory* is a symmetric monoidal category $(\mathcal{C}, \otimes, \mathbb{1})$ together with two designated maps $q_S: S \rightarrow \mathbb{1}$ and $w_S: \mathbb{1} \rightarrow S$, satisfying standard monoidal naturality conditions. Objects in $\mathcal{C}$ are interpreted as system types, and morphisms as physical processes. Morphisms of the form $\mathbb{1} \rightarrow S$ are called *states*, while morphisms $S \rightarrow \mathbb{1}$ are called *effects* and morphisms $\mathbb{1} \rightarrow \mathbb{1}$ are called *scalars*. The designated maps q and w will be referred to as the *discarding effect* and the *noise state* respectively. A morphism $f: S \rightarrow S'$ is *causal* if it preserves the discarding effect,

$$q_{S'} \circ f = q_S,$$

and similarly *cocausal* if it preserves the noise state. We will write $\text{St}(S)$ for the collection of states of S .

Remark 2.1. Sometimes it is natural to require $\mathcal{C}$ to be a symmetric monoidal *dagger* category, which is interpreted as modeling a time-reversal operation.

Example 2.2. The standard example of a process theory is the category of classical probabilistic processes. This is the category Class of finite sets, where a morphism $S \rightarrow S'$ assigns to an element $s \in S$ an 'unnormalized probability distribution' of elements in S' , which are maps

$S \times S' \longrightarrow \mathbb{R}^+$. Composition of two maps $f: S \times S' \longrightarrow \mathbb{R}^+$ and $g: S' \times S'' \longrightarrow \mathbb{R}^+$, is defined by

$$\begin{aligned} f \circ g: S \times S'' &\longrightarrow \mathbb{R}^+ \\ (s, s'') &\longmapsto \sum_{s' \in S'} f(s, s')g(s', s''). \end{aligned}$$

The monoidal structure is the cartesian product, meaning the unit $\mathbb{1}$ is the singleton set. Product of morphisms is given by

$$f \otimes g(s, s'')(s', s''') = f(s, s')g(s'', s''').$$

The discarding process q_S is the unique process with $q_S(s) = 1$ for all $s \in S$, meaning a process is causal whenever it is stochastic. The noise state is the uniform probability distribution with $w_S(s) = \frac{1}{|S|}$. This example has a natural time reversal operation, given by $f^\dagger(s, s') = f(s', s)$.

Let $(\mathcal{C}, \otimes, \mathbb{1}, q)$ be a process theory, and S an object in $\mathcal{C}$. A *decomposition* of S is a pair of objects $A, B \in \mathcal{C}$ together with a causal isomorphism $\psi: S \simeq A \otimes B$. We denote decompositions by (A, B, ψ) , or sometimes simply by (A, B) .

Any object S has a decomposition $(S, \mathbb{1})$, which we call the *trivial* decomposition. Two decompositions (A, B, ψ) and (A', B', ψ') are *equivalent* if there are causal isomorphisms $f: A \simeq A'$ and $g: B \simeq B'$, such that

$$\psi' = (f \otimes g) \circ \psi.$$

The *complement* decomposition of (A, B) is the same decomposition, just in the opposite order; it is denoted $(A, B)^\perp = (B, A)$. If two decompositions are equivalent, then so are their complements.

We define $\mathbb{D}(S)$ to be the set of equivalence classes of decompositions of S . The operation $(-)^\perp$ acts on $\mathbb{D}(S)$, and we define a *decomposition set* of S to be a subset $\mathbb{D} \subseteq \mathbb{D}(S)$ that contains the trivial decomposition and is closed under $(-)^\perp$.

Given a decomposition set $\mathbb{D}$ of S , and a decomposition $(A, B, \psi) \in \mathbb{D}$, we define the *restricted* decomposition set $\mathbb{D}|_A$ to be the set of decompositions of A that can be extended to a decomposition of S in $\mathbb{D}$. Furthermore, we define the *restriction* of a state $s: \mathbb{1} \longrightarrow S$ to A to be the state defined by

$$s|_A = (\text{Id}_A \otimes q_B) \circ \psi \circ s: \mathbb{1} \longrightarrow A,$$

and similarly for B . Intuitively, we identify S with its decomposition, and then discard the factor we don't want to consider. By functoriality and naturality, such restrictions depend only on the equivalence class in $\mathbb{D}(S)$, and not on the particular representative.

2.2 System types and directional cuts

To define generalized integrated information theories, we need a notion of systems to study – defined as follows. A *system* is a triple $(S, \mathbb{D}, T)$, where $S \in \mathcal{C}$, $\mathbb{D}$ is a decomposition set of S and $T: S \longrightarrow S$ is a causal process that is interpreted as *time evolution*. The most basic example is the *trivial* system $(\mathbb{1}, \{\mathbb{1} \otimes \mathbb{1}\}, \text{Id}_{\mathbb{1}})$.

A state of a system $(S, \mathbb{D}, T)$ is a state of S . Given such a state s and a decomposition $(A, B) \in \mathbb{D}$, [CITE](#) constructs a *subsystem* $(A, \mathbb{D}|_A, T|_A)$, where the restricted time evolution is defined by the composition

$$A \xrightarrow{1 \otimes s|_B} A \otimes B \xrightarrow{T} A \otimes B \xrightarrow{1 \otimes q_A} A.$$

These subsystems are again independent of the class representative in $\mathbb{D}$.

Given a system $(S, \mathbb{D}, T)$ and a decomposition $(A, B, \psi) \in \mathbb{D}$, one needs to be able to form systems that disconnects certain causal connections in the total system. IIT 3.0 [CITE](#) does this by introducing *cut systems*, that directionally disconnects causal flow from the two parts. IIT 4.0 [CITE](#) does this in a slightly different manner, by replacing parts of the inputs of the time evolutions with independent noise. We have included versions of IIT 4.0's disconnections in arbitrary system states for reference and intuition, but we remark that the actual implementation of such disconnections does not matter much, as long as they are functorial.

The B -cut replaces the contribution of B to the time evolution with the noise state, meaning that only the contribution from A matters. It can formally be described as follows:

$$T^{\psi, \leftarrow} : A \otimes B \xrightarrow{1 \otimes w_B q_B} A \otimes B \xrightarrow{T} A \otimes B$$

In spirit this forces the time evolution to retain all self-influences from A , and from A to B , but removes all influences from B to A . The arrow in the notation is meant to denote the direction that is removed. Similarly, the A -cut is defined by

$$T^{\psi, \rightarrow} : A \otimes B \xrightarrow{w_A q_A \otimes 1} A \otimes B \xrightarrow{T} A \otimes B,$$

and the A - B -cut by

$$T^{\psi, \leftrightarrow} : A \otimes B \xrightarrow{1 \otimes w_A \otimes w_B \otimes 1} A \otimes B \otimes A \otimes B \xrightarrow{T \otimes T} A \otimes B \otimes A \otimes B \xrightarrow{1 \otimes q_B \otimes q_A \otimes 1} A \otimes B.$$

The A - B -cut corresponds to Tull–Kleiner's bidirectional cut, where one evolves the system as if A and B were completely disconnected from each other.

We will use the notation $T^{\psi, \delta}$, with $\delta \in \{\leftarrow, \rightarrow, \leftrightarrow\}$ to denote a general disconnection based on a decomposition $\psi : S \simeq A \otimes B$. As we did above with the restricted time evolution, we also get definitions of restricted cuts: $T|_A^{\psi, \delta}$.

3 Integrated cohomology and obstruction classes

In this section we present the more novel parts of this paper, which is a cohomology theory for IIT systems. We then build an obstruction theory for decomposing a physical system based on this cohomology theory, and show that it detects the existence of integrated information. Lastly we extend this to naturally define higher order integration, often referred to as higher *synergy*.

Remark 3.1. Our setup is formally analogous to Abramsky–Brandenburger's sheaf theoretic description of contextuality, see [CITE](#) and [CITE](#), but differs in the added dynamics of the systems. However, we feel that the similarities are enticing, and that the possible interactions are worth further studies.

Given a system type $(S, \mathbb{D}, T)$ and a global state $s: \mathbb{1} \rightarrow S$, we define the category $\text{Sub}_s(S)$ of subsystems of S to be the poset category defined by the poset structure in [CITE](#). In other words, $\text{Sub}_s(S)$ is the category consisting of subsystems as objects, and morphisms being single comparison maps by the preorder structure.

The idea is then as follows:

1. define a presheaf $\mathcal{F}$ on $\text{Sub}_s(S)$ that describes the cause-effect dynamics of subsystems of S , where the restriction maps describe subsystem dynamics;
2. define the integrated cohomology of S to be the presheaf-cohomology of the nerve of $\text{Sub}_s(S)$ valued in $\mathcal{F}$;
3. define natural obstruction classes in $H^1(N^\bullet(\text{Sub}_s(S)); \mathcal{F})$;
4. prove that these obstruction classes detect the existence of integrated information in S .

There are several possible choices for what presheaf $\mathcal{F}$ to use, depending on how categorical or how system specific one wants to be. If one focuses on actual IIT 4.0 constructions, then perhaps a probability based approach using intrinsic information and intrinsic difference would be a good strategy. We have, however, opted for a more categorical and general approach. This is in part to give integrated information more functorial properties, as discussed by Tull–Kleiner in [CITE](#).

As we want to have access to the time evolutions T and $T^{\psi, \delta}$ for subsystems, it is natural to consider the *endomorphisms* as a natural starting position for the presheaf. As we need to be able to understand the *difference* between T and $T^{\psi, \delta}$, we also need access to a subtraction operation. The standard way to do this is to define the presheaf on objects by

$$\begin{aligned} \mathcal{F}: \text{Sub}_s(S)^{\text{op}} &\rightarrow \text{Ab} \\ A &\mapsto \mathbb{Z}[\text{End}(A)] \end{aligned}$$

in other words it is the free abelian group on $\text{End}(A)$. Given a subsystem $U \leq A$, we define for any endomorphism $f: A \rightarrow A$ a restriction map

$$\rho_U^A(f) = (\text{Id}_U \otimes q_V) \circ f \circ (\text{Id}_U \otimes w_V),$$

where V is the complement of U , i.e. the object such that $A \simeq U \otimes V$, which exists by the definition of a subsystem. We then linearly extended this to $\mathbb{Z}[\text{End}(A)]$, making $\mathcal{F}$ a presheaf.

Lemma 3.2. *The presheaf $\mathcal{F}$ is a sheaf.*

Prove this fact? Is it needed?

Remark 3.3. One could also naturally choose to use the actual state $s|_V$ instead of the noise state w_V in the definition of the restriction maps. Doing this would mean that the obstruction theory detects state-dependent integration, and not the intrinsic integration of the system. We believe that studying what happens upon making this change is an interesting avenue of future research.

Given a system state $(S, \mathbb{D}, T)$ and state $s: \mathbb{1} \rightarrow S$, we define the *integrated cohomology* of $(S, \mathbb{D}, T)$ to be the $\mathcal{F}$ -valued presheaf cohomology of the nerve of $\text{Sub}_s(S)$:

$$H_{int}^k(S) := H^k(N^\bullet(\text{Sub}_s(S)); \mathcal{F}).$$

Remark 3.4. It is important to note that this is not simply the topological nerve cohomology of $\text{Sub}_s(S)$, as this will almost always be trivial. The important part here is the use of the presheaf $\mathcal{F}$, which makes this a highly non-trivial cohomology theory.

The intuition is as follows: The nerve of $\text{Sub}_s(S)$ gives us a geometric construction of *where* integrated information can be tested. It consists formally of chains of substates of S of all lengths, with topological structure that understands how these chains are related and connected. The presheaf gives us *what* is measured on every level – in our case the system’s causal dynamics. The cohomology gives us information about which local dynamics that can be glued together to form a global dynamic. If gluing fails, then there is some level of the system where its global dynamics cannot be described in terms of local dynamics, which is the essence of the system having integrated information.

The *decomposition obstruction classes* of S are defined as the cohomology class of the cochains

$$\Delta^{\psi,\delta}(A) = T|_A - T|_A^{\psi,\delta}$$

for all substates A , decompositions ψ and directions $\delta \in \{\rightarrow, \leftarrow, \leftrightarrow\}$.

Remark 3.5. The reader paying minute attention has perhaps noticed that mechanisms and purviews have yet to be featured in this framework. Our approach to measuring IIT is structurally somewhat different, as instead of starting with mechanisms and purviews and then aggregate local information upwards, we have the global structure as a reference and let mechanisms and purviews be derived observables. Hence they do not arise as primitive data, but as local test probes for evaluating the obstruction classes. We will, however, need to define and use these to relate our obstruction classes to the integrated information of S , as we will do in the following section.

3.1 Obstruction classes and generalized IIT

The goal of this section is to prove the following theorem.

Theorem 3.6. *Let $(S, \mathbb{D}, T)$ be a system state and $s: \mathbb{1} \rightarrow S$ a state of S . If there is no integrated information in S , then all of the decomposition obstruction classes vanish. In other words, $\Phi(S, s) = 0 \implies [\Delta^{\psi,\delta}(A)] = 0$ for all A , ψ and δ . In particular, the existence of non-vanishing obstruction classes implies $\Phi(S, s) > 0$, and hence that the classes $[\Delta^{\psi,\delta}(A)]$ form obstructions to the decomposition of causal structures.*

In order to do so we need to be able to compute and understand Φ . We will use the method of *generalized IIT’s* as developed by Tull–Kleiner in [CITE](#), also formalized in the categorical setting in [CITE](#). This ensures that our theory works not only for classical IIT, but also quantum IIT and other variants that one would like to study, [CITE](#).

A *mechanism-purview pair* is a pair of subsystems of a given system $(S, \mathbb{D}, T)$, depending on the same global state $s: \mathbb{1} \rightarrow S$. Intuitively a mechanism describes the subset of S that we are “letting make a difference”, and the purview describes the subset of S that is affected.

Given a mechanism $(M, \mathbb{D}|_M, T|_M)$ and a purview $(P, \mathbb{D}|_P, T|_P)$, we define *causes* and *effects* to be processes

$$cau: M \rightarrow P \text{ and } eff: M \rightarrow P.$$

These capture the way the previous and next state of P is constrained by the restricted state $s|_M$. We require that both cau and eff send causal states to causal states. When the mechanism is the unit $\mathbb{1}$, then a cause $\mathbb{1} \rightarrow P$ is called an *unconstrained case*, and is denoted $ucau$.

One example of such a cause process is

$$cau: M \xrightarrow{1 \otimes w_{M'}} M \otimes M' \simeq S \xrightarrow{T} S \simeq P \otimes P' \xrightarrow{1 \otimes q_{P'}} P,$$

where M' and P' are the complements in the decompositions defined by M and P respectively.

Given a state s and a mechanism-purview pair M, P , the associated *cause repertoire* is defined to be the state

$$cau(s, M, P) = \mathbb{1} \xrightarrow{s|_M} M \xrightarrow{cau \otimes ucau} P \otimes P' \simeq S.$$

One defines an *effect repertoire* $eff(s, M, P)$ similarly.

We also need to decompose causes and effects. Given a decomposition (M_1, M_2) of a mechanism M , and a decomposition (P_1, P_2) of a purview P , the *decomposed* cause is the process

$$cau_{M_1, M_2}^{P_1, P_2}: M \simeq M_1 \otimes M_2 \xrightarrow{cau \otimes cau} P_1 \otimes P_2 \simeq P,$$

which again similarly defines the *decomposed* effect. The *decomposed* cause and effect repertoires are then defined similarly to the normal ones, by replacing cau and eff with their decomposed variants in the definitions. We denote these states by $cau_{M_1, M_2}^{P_1, P_2}(s, M, P)$ and $eff_{M_1, M_2}^{P_1, P_2}(s, M, P)$ respectively.

For a generalized IIT, one needs to be able to compare causes and effects along different substructures. This is done by assuming that there is a distance function on the states of S : $d(-, -): \text{St}(S) \times \text{St}(S) \rightarrow \mathbb{R}^+$. As in [CITE](#), we do not require this to be a metric, though it usually will be in examples. In classical IIT, this metric is the Wasserstein metric, while in quantum-classical IIT it is the metric induced by the norm of the system C^* -algebra.

We now have enough data to define the IIT-algorithm, as presented in [CITE](#) and [CITE](#). The idea is to first define the integration level of cause and effect repertoires by

$$\varphi(cau(s, M, P)) := \min_{(M_1, M_2), (P_1, P_2)} d(cau(s, M, P), cau_{M_1, M_2}^{P_1, P_2}(s, M, P)),$$

and then for each M define the *core cause* purview P^{cc} that give the maximal integration. Similarly one defines the integration of effect repertoires, and use this to define for each mechanism the *core effect* purview P^{ce} that maximizes the integration. The integration of a mechanism M is then defined as

$$\varphi(M) := \min(\varphi(cau(s, M, P^{cc})), \varphi(eff(s, M, P^{ce}))).$$

The core cause repertoire, core effect repertoire and integration defines the *concept* of M – see [CITE](#) for details. Allowing M to vary, defines the *Q-shape* of the state s . The collection of all *Q-shapes* defines the *experience space* $\mathbb{E}(S)$ of the system S .

By minimizing over all possible decompositions, one can similarly define the integration of a *Q-shape*. The subsystem with maximally integrated *Q-shape* is the system's *major complex*,

which by definition defines an element $\mathbb{E}(S, s)$ in the experience space $\mathbb{E}(S)$. IIT claims that this element is the particular conscious experience of the system S in the state s , and the associated Φ -value is defined as the size of this conscious experience: $\Phi(S, s) = \|\mathbb{E}(S, s)\|$.

If this Φ -value is zero, then it means that for all mechanisms M , the integration $\varphi(M)$ is zero, meaning that for all purviews P there is a decomposition with the same cause and effect repertoire. In principle, decomposing the system does not change its cause and effect repertoires. We now use this to prove our main result.

Proof of Theorem 3.6. The assumption that $\Phi(S, s) = 0$ implies by definition that for all mechanisms M , we have

$$\varphi(M) = \min(\varphi(\text{cau}(s, M, P^{cc})), \varphi(\text{eff}(s, M, P^{ce}))) = 0.$$

In particular, for all mechanism-purview pairs M, P , there exists decompositions $M \simeq M_1 \otimes M_2$ and $P \simeq P_1 \otimes P_2$ such that either

$$\text{cau}(s, M, P) = \text{cau}_{M_1, M_2}^{P_1, P_2}(s, M, P),$$

or

$$\text{eff}(s, M, P) = \text{eff}_{M_1, M_2}^{P_1, P_2}(s, M, P).$$

The goal is to exploit this fact to show that all obstruction classes are coboundaries. To accomplish this, we fix a decomposition $\psi: S \simeq A \otimes B$. If the cut direction is $\rightarrow$ $\square$